ETL Walmart Midterm Project

Mark Chernov, Michael Tague,
Jun Seo Lee

The Team



Mark Chernov

Mathematics

Class of ‘28



Michael Tague

Architecture

Class of ‘28



Jun Seo Lee

Commerce

Class of ‘27

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Workflow

Data Cleaning

- **Process across the 5 files**
  - Drop duplicates, identify basic convention errors, wrangle with larger errors/mismatched data
  - Turned all numbers into numeric values that pandas could read
  - Used functions to apply data cleaning principles across multiple files that share similar informations
- **Main challenges**
  - Supplier mix-up in the Products file
  - Using sqlite3 to evaluate the Inventory and Sales file
  - Working with thousands of rows of Transactions data

Weather API and Joins

- **Data collection pipeline**
  - Requests to Open-Meteo using location coordinates
  - Retry logic system is in place to prevent actions from being taken with insufficient data.
- **How the data is organized**
  - When JSON data is retrieved, it is uploaded to a dataframe where it is then renamed, timestamped, and tagged with a source column.
- **Joining the tables**
  - Time discrepancies between UTC and Eastern timestamps were normalized to midnight times allowing for a precise, audited left join on Store ID and date that saw no dropped rows.

Data Analysis

- **Prepared the analysis dataset**
  - 24,521 dated sales records across 12 stores, totaling 47,897 recorded units.
  - Removed exact duplicates, retained positive quantities, and excluded 881 positive-sale records with unusable dates.
- **Established consistent comparison windows**
  - Baseline: September 1–15, 2024.
  - Pre-landfall comparison: September 19–25.
  - Compared average daily units because the windows contain different numbers of days.
- **Investigated five business questions**

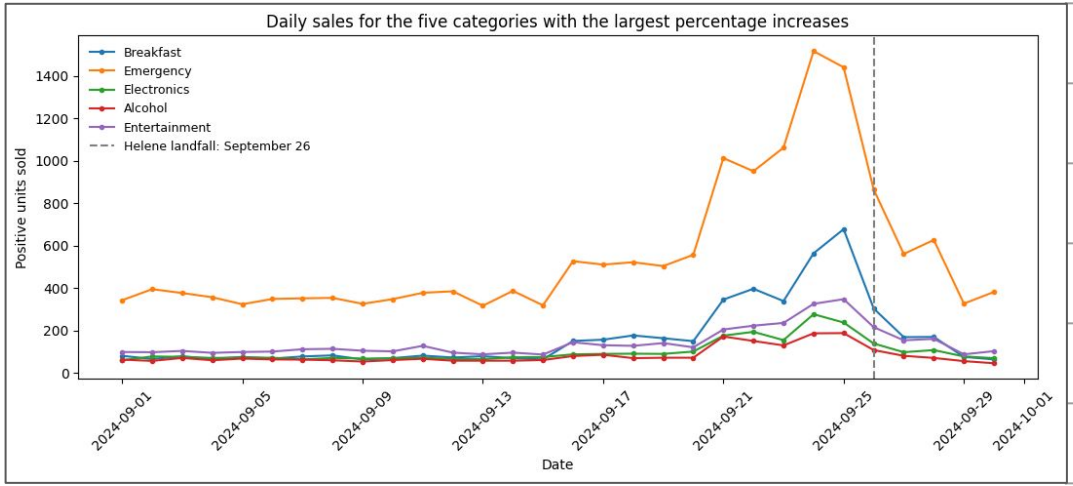
Charts and Presentation

- **Generating the charts**
  - Plotting time-series overlays to compare daily unit sales against baseline averages for high-velocity items like bottled water, batteries, and Pop-Tarts.
  - Using bar charts when mapping categories or store locations
- **Structuring the findings**
  - Making each chart informative as possible while still making it concise and readable within a small area
  - Answering each question in just a few sentences so that dense text on the screen did not make for a overwhelming reading experience

02

Data Analysis

Question 1: Demand Surge Identification



Category	Increase in average daily units
Breakfast	**414.36%**
Emergency	**184.25%**
Electronics	**150.27%**
Alcohol	**125.66%**
Entertainment	**124.97%**

Caption: The graph shows multiple line charts of daily positive-unit sales for 5 product category with the largest percent increase in sales and an annotation marking the landfall date of Hurricane Helene.

Largest Increase

The Breakfast category had the largest percentage increase from the baseline(September 1-15) to comparison(September 19-25) window, where the average daily sales rose from 73.27 to 376.86 units, a 414% increase

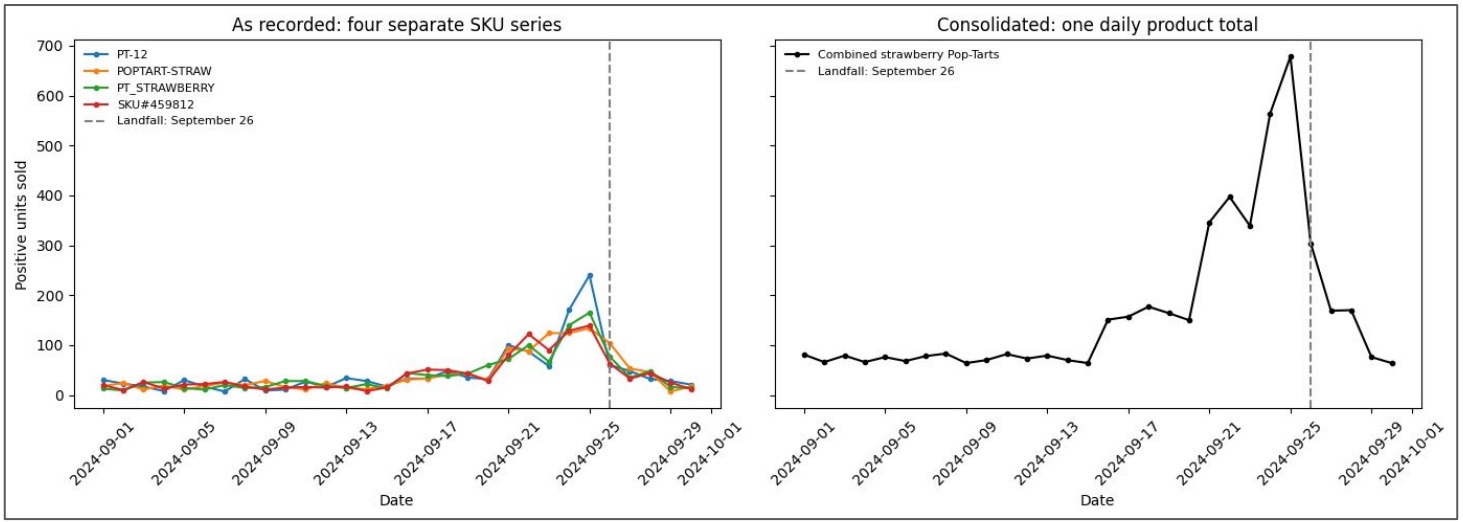
Percent vs Quantity

Though Emergency had the highest daily volume of any category, at a little over 1 million units sold during the comparison window, Breakfast saw a larger relative growth in purchase volume compared to its baseline window than Emergency did

Data points

Our analysis comes from around 24,500 dates, and positive quantity sales records. Duplicates were removed and any non positive quantities were also dropped.

Question 2: The Pop-Tarts Effect (SKU Consolidation)



Caption: Figure 1, the left chart, shows multiple line charts of Pop tart sales numbers under different SKUs. Figure 2, the right chart, shows the sum of all Pop Tart sales for each day, merged into one SKU id.

Merging

Consolidating four strawberry Pop-Tarts identifiers revealed 5,005 recorded units during September.

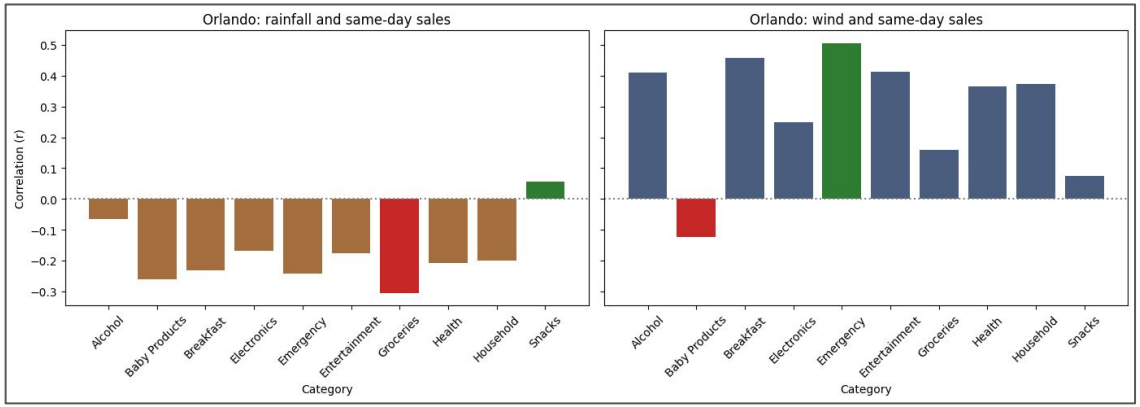
Peak

Combined daily sales peaked at 678 units on September 25, one day before landfall.

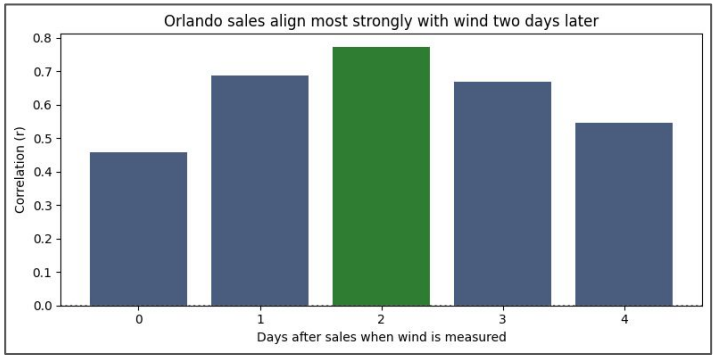
Popularity

Planning from the combined product family gives buyers a fuller demand picture while preserving the original unit total.

Question 3: Weather Correlation Analysis



Caption: Figure 1, left chart, displays the correlation between correlation between daily category sales and rainfall measurement. Figure 2, right chart, displays correlation between daily category sales and each weather measurement. There exists a red and green bar demonstrating greatest and lowest correlation for each figure.



Caption: The graph shows a correlation between overall sales data for Orlando and tested the lag from zero to four days while selecting the highest correlation, so the result should be treated as exploratory.

Wind correlation

Same-day wind correlation: r = 0.458, using 30 paired days.

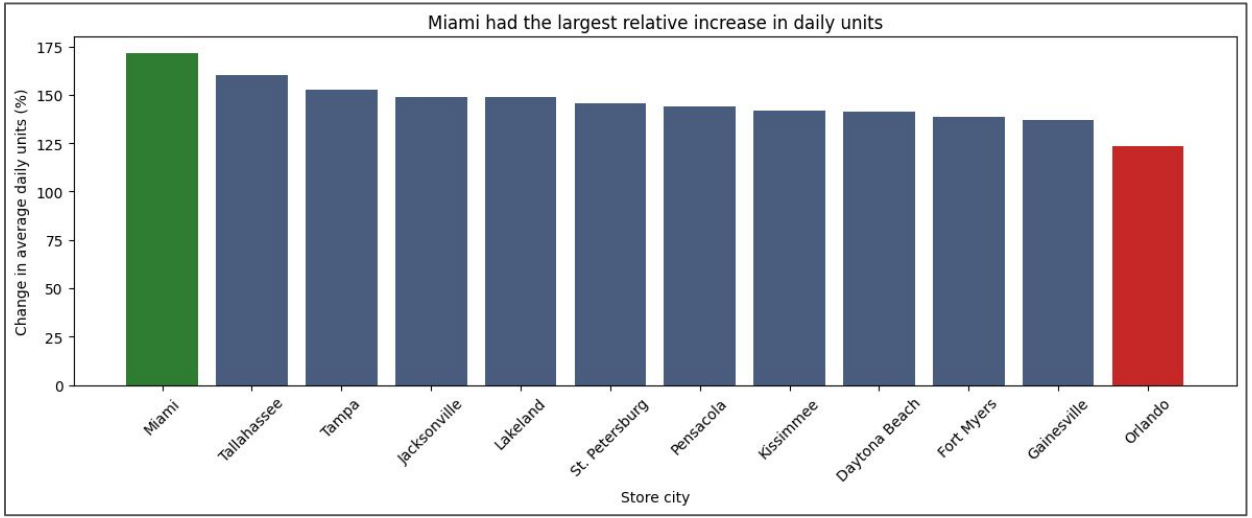
2 Days later

Sales paired with wind two days later: r = 0.773, using 28 paired days.

Popularity

Across Orlando categories, Emergency had the strongest same-day wind correlation: r = 0.506, using 30 paired days.

Question 4: Store-Level Geographic Patterns



Caption: The graph represents a bar chart and displays the difference in relative increase in daily units for each store, with a green bar marking the store with the highest relative increase and a red bar marking the lowest relative increase.

Growth

All 12 stores increased.

- Miami had the largest relative increase (171.44%).
- Orlando had the smallest (123.42%).

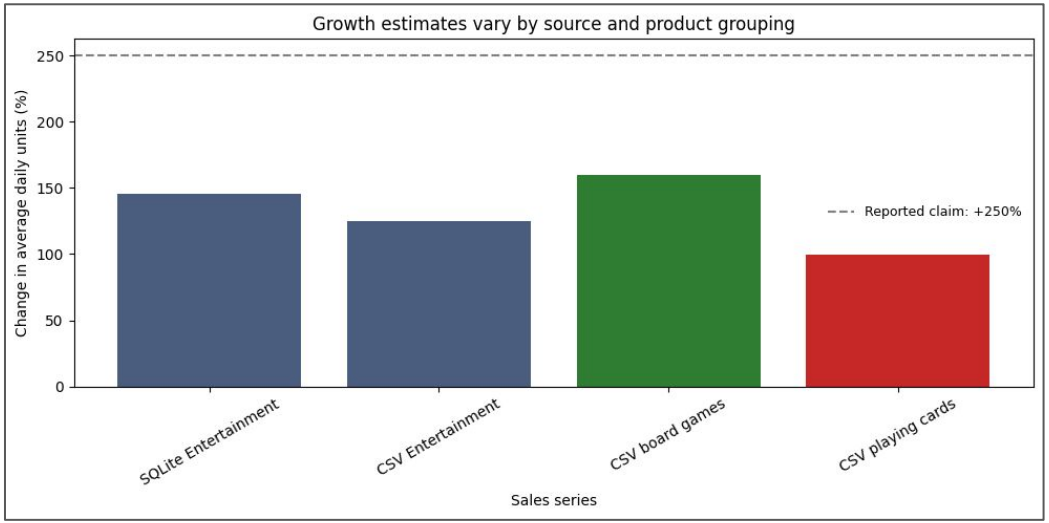
Best Average

Lakeland had the highest comparison-period average: 239.43 units per day.

Peaks

Every store reached its monthly peak on September 24 or 25, one or two days before landfall.

Question 5: Signal vs. Artifact



Caption: The graph represents a bar chart and displays the estimated growth of various products separated into shared categories with the highest expected growth bar marked in green and the lowest expected growth bar marked in red. It also contains an annotation with a horizontal marker of the reported 250% sales increase, which no bar demonstrates.

Sales growth

Recorded Monopoly sales increased 159.94%. Entertainment also includes playing cards, so category growth does not measure board games alone.

Disconnect in data

Monthly Entertainment totals disagree: 24,070 units in SQLite versus 4,250 in the cleaned CSV, despite matching all 360 store-days.

Inventory

Inventory records cover 218 of 360 possible store-days, and 45 records lack order quantities.

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Reflection

RQ1: Data Quality Impact

Quality Issue

Suppliers were mismatched with products in the Product Catalog

Specific Example

Pop-Tarts were assigned to the supplier Roku

Implications

Mismatched information would have been sent to distribution centers, causing confusion over what product to stock and from which supplier

Solution

Noting which suppliers sell which products in real life and translating that information into a dictionary map that revises the catalog to correctly match supplier with corresponding product

RQ2: ETL Trade-offs

Keeping stores

We manually supplied missing location details and selected local weather timezones to retain affected stores.

Dataset-specific rules

Store-specific rules and a distribution-center lookup dictionary supported weather retrieval for the supplied dataset.

Easier maintenance

A separate location-cleaning function would make the pipeline easier to maintain as new locations are added.

Alternative approach

Dropping incomplete stores would simplify retrieval but exclude their sales from weather comparisons, potentially changing the findings.

RQ3: Pipeline Trust

Reliable inputs

Pipeline errors can undermine an otherwise correct analysis.

Manual location data

Weather retrieval relied on supplied coordinates and location-specific timezone rules.

Consistent join keys

Store IDs and dates needed consistent formatting to match sales with weather.

Validation checks

We excluded undated sales before daily analysis and checked that the join preserved eligible rows and unit totals.

RQ4: Business vs. Data

Sales increase

Board game sales increased 160%, supporting consideration of extra stock before future events.

Conflicting data

Conflicting sales totals and incomplete inventory records limited confidence

Real risks

Balancing excess inventory against missed sales and reduced product availability

Recommendations

Check each store’s stock and open orders before replenishing. With more time, we would reconcile both sales sources and compare more historical data.

RQ5: Team Collaboration

Mark

Mark asked useful questions, persevered through join issues, and completed the API calls and joins.

Michael

Michael carefully cleaned the data, standardized SKUs, and tracked records to preserve rows and build a reliable foundation for analysis.

J.S.

JS provided leadership, programming guidance, and initiative, while keeping the project on schedule and helping shape the presentation.

Teamwork & Next Steps

We divided responsibilities, met regularly, and helped each other solve problems. With another week, we'd improve reuse and automate store-location lookups.

Thanks

Questions?

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